

49. Which statement best captures the role and limitations of surrogate losses for binary classification?

1 Point

- ☐ Minimizing any convex surrogate is exactly equivalent to minimizing 0-1 loss for finite samples.
- ☐ The unscaled logistic loss $\log(1 + e^{-u})$ is a pointwise upper bound on $\mathbb{1}\{u < 0\}$ for every u .
- ☐ Convex surrogate losses make optimization tractable and can upper-bound 0-1 loss when properly chosen or scaled; however, surrogate minimization is still a proxy and not literally identical to 0-1 minimization.
- ☐ Surrogate losses are used only because they are discontinuous like 0-1 loss.

50. Which comparison between empirical 0-1 risk minimization over halfspaces and surrogate-risk minimization is most accurate?

1 Point

- ☐ 0-1 risk is smooth but nonconvex, while surrogate losses are discontinuous but convex.
- ☐ 0-1 halfspace ERM is discontinuous and computationally difficult in general; convex surrogate objectives replace it with tractable proxy problems in the score or margin.
- ☐ Surrogate losses remove the need for test sets because they exactly equal true risk.
- ☐ Logistic regression minimizes squared loss on labels encoded as $\{-1, 1\}$.

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45. Two binary classification distributions have the same decision boundary $\{x : \eta(x) = 1/2\}$, but in one distribution $\eta(x)$ changes very sharply across the boundary, while in the other it changes gradually. Which statement is correct?

1 Point

- ☐ They can have different Bayes risks even with the same boundary, because the amount of posterior overlap/confusion differs.
- ☐ They must have identical Bayes risk because the classifier labels the same regions.
- ☐ The gradual case must have lower Bayes risk because it is smoother.
- ☐ Bayes risk is determined only by the class priors, not by posteriors near the boundary.

46. Consider the binary classifier

$$f_w(x) = \text{sign}(w^T \phi(x)), \quad \phi(x) = (1, x_1^2, x_2^2), \quad w = (-1, 1, 1).$$

1 Point

Ignoring the conventions on the boundary, which decision region is labeled +1?

- ☐ The inside of the unit circle, $x_1^2 + x_2^2 < 1$.
- ☐ The outside of the unit circle, $x_1^2 + x_2^2 > 1$.
- ☐ The half-space $x_1 > x_2$.
- ☐ The two cones $|x_1| > |x_2|$.

47. Why do binary linear classifiers often model a real-valued score $g_w(x) = w^T \phi(x)$ and then output $\text{sign}(g_w(x))$?

1 Point

- ☐ Because the labels -1 and 1 have ordinal distances that must be preserved.
- ☐ Because the score itself is a probability.
- ☐ Because sign functions are convex.
- ☐ Because real-valued scores can model logits or separating surfaces conveniently, while the final class label is obtained by thresholding the score.

48. Let the margin of an example be $u = y w^T \phi(x)$. Which statement about common surrogate losses is correct?

1 Point

- ☐ Hinge loss $\max(0, 1 - u)$ is zero only when $u \geq 1$, while logistic $\log(1 + e^{-u})$ and exponential e^{-u} remain positive but decay as u increases.
- ☐ All three losses are zero whenever $u > 0$.
- ☐ Logistic loss increases as the positive margin grows.
- ☐ Exponential loss is discontinuous at $u = 0$.

41. For classification with 0-1 loss, what is the Bayes optimal classifier?

1 Point

☐ $f^*(x) \in \arg \max_y \mathbb{P}(Y = y \mid X = x)$.
 ☐ $f^*(x) = \mathbb{E}[Y \mid X = x]$ for all label sets.
 ☐ $f^*(x)$ is the class with smallest prior probability.
 ☐ $f^*(x)$ is any constant classifier.

42. In binary classification with labels $\{-1, 1\}$, suppose $\mathbb{P}(Y = 1 \mid X = x) = 0.7$. What does the Bayes classifier predict at x , and what is the pointwise probability of error?

1 Point

☐ Predict -1, error 0.3.
 ☐ Predict 1, error 0.3.
 ☐ Predict 1, error 0.7.
 ☐ Predict either class, error 0.3.

43. Let $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$ with $\eta(x) \in (0, 1)$. Which equivalence is correct for binary Bayes classification away from tie conventions?

1 Point

☐ $\text{sign}(2\eta(x) - 1) = \text{sign}(\eta(x)(1 - \eta(x)))$.
 ☐ $\text{sign}(2\eta(x) - 1) = \text{sign}(\log(1 - \eta(x)))$.
 ☐ $\text{sign}(2\eta(x) - 1) = \text{sign}\left(\log \frac{\eta(x)}{1 - \eta(x)}\right)$.
 ☐ $\text{sign}(2\eta(x) - 1) = \eta(x)$.

44. For binary classification, the pointwise Bayes error at x is $\min\{\eta(x), 1 - \eta(x)\}$. Where is this pointwise error maximized?

1 Point

☐ Where $\eta(x) = 0$.
 ☐ Where $\eta(x) = 1$.
 ☐ Far away from the decision boundary.
 ☐ At a posterior (ie. $\eta(x) = 1/2$), where the error is $1/2$.

37. Consider square-loss regression with fixed feature map $\phi(x) = (1, x, x^2)$. Which statement is correct?

1 Point

- ☐ The optimization is nonlinear and nonconvex because x^2 appears.
- ☐ The model is linear in the raw input x .
- ☐ The model is nonlinear in x but linear in w , so the least-squares objective remains a convex quadratic in w .
- ☐ The normal equations cannot be formed for polynomial features.

38. Suppose the current restricted regression model gives

1 Point

$Y = w^T \phi_0(X) + N$, $E[\phi_0(X)N] = 0$,
but $E[N | X]$ is not zero. What is the effect of adding a new feature $\phi_1(X)$ that is strongly aligned with $E[N | X]$?

- ☐ It cannot affect model error because N is already orthogonal to $\phi_0(X)$.
- ☐ It necessarily increases both model error and estimation error.
- ☐ It can move predictable structure from the residual into the modeled component, potentially reducing model error, though it may increase estimation error.
- ☐ It makes the residual independent of X automatically.

39. Why does universal approximation not by itself guarantee practical learnability from finite samples?

1 Point

- ☐ A universal approximator may require a very large number P of parameters/features, and estimation error bounds often scale at least like P/m or worse.
- ☐ Universal approximation applies only to classification, not regression.
- ☐ If a model is universal, empirical risk is always equal to true risk.
- ☐ Universal approximation eliminates noise from the data-generating process.

40. A modelling strategy based on polynomial feature maps may suffer from computational challenges and the curse of dimensionality because the number of basis terms can grow very rapidly. Which statement best describes two later modelling directions designed to address limitations of plain basis expansion?

1 Point

- ☐ Kernel-based modelling is introduced because ordinary linear modelling has no computational or estimation challenges.
- ☐ Neural network modelling is exactly the same as taking a fixed linear combination of a prechosen basis.
- ☐ Kernel methods are unrelated to the curse of dimensionality discussed for polynomial feature maps.
- ☐ Kernel-based modelling can alleviate computational issues and address the curse of dimensionality, while neural-network modelling uses composition rather than only combination, with modularity as a design advantage.

33. A degree- d monomial in n input variables has the form

$$x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}, \quad \alpha_i \geq 0, \quad \alpha_1 + \cdots + \alpha_n = d.$$

How many such degree- d monomial exponent patterns are there?

- ☐ $\binom{n}{d}$.
- ☐ n^d .
- ☐ $\binom{n+d-1}{d}$.
- ☐ $d+n-1$.

34. Why can sufficiently rich linear modelling with basis functions act as a universal approximation strategy?

- ☐ Because any finite training set can be memorized, independent of the chosen basis.
- ☐ Because if the chosen basis covers the relevant function family, then by taking enough basis terms, linear combinations of those elementary functions can approximate or represent very rich target functions.
- ☐ Because increasing the number of basis terms always decreases test error.
- ☐ Because the empirical correlation matrix becomes exactly the identity when enough basis functions are used.

35. Which comparison of basis choices and their inductive biases is most accurate?

- ☐ Monomial, Fourier, and Legendre bases encode exactly the same ordering of functions.
- ☐ A Fourier basis primarily orders functions by polynomial degree.
- ☐ A monomial basis primarily encodes frequency bias.
- ☐ Monomial bases naturally organize by degree, Fourier bases by frequency, and Legendre-type bases by orthogonal polynomial shape; the number of retained terms controls bias strength.

36. Let $P(p, D) = \binom{p+D}{D}$ be the number of monomials of total degree at most D in p variables. Suppose $D = \alpha p$ for a fixed $\alpha > 0$. Which statement best describes the implication for sample complexity in underparameterized least squares with noise variance σ^2 ?

- ☐ By Stirling's approximation, $P(p, \alpha p)$ grows exponentially in p , so a lower-bound term like $\sigma^2 P(p, D)/m$ requires m to grow comparably to keep estimation error controlled.
- ☐ $P(p, \alpha p)$ is always linear in p , so no curse of dimensionality arises.
- ☐ Universal approximation implies m can stay fixed as p grows.
- ☐ Since the model is linear in parameters, the number of monomials has no effect on estimation error.

29. A test set is intended to estimate the true risk of a trained predictor under a target distribution $p^*(x, y)$. Suppose the trained predictor f is evaluated on a large independent dataset drawn from a different distribution $q(x, y) \neq p^*(x, y)$. What does the average loss on this dataset estimate? 1 Point

- ☐ It estimates the training error, because f was learned from the training set.
- ☐ It estimates the true risk under p^* as long as the dataset is large.
- ☐ It estimates the expected loss under q , not necessarily the true risk under p^* .
- ☐ It estimates the Bayes optimal risk, because the dataset is independent.

30. A sequence of increasingly rich models gives the following approximate errors: 1 Point

Model	Estimation error	Model error	Training error trend
$\mathcal{M}_1$	1	8	high
$\mathcal{M}_2$	3	3	lower
$\mathcal{M}_3$	9	0	lowest

Which model should be preferred if the goal is low generalization error?

- ☐ $\mathcal{M}_1$ because it has the smallest estimation error.
- ☐ $\mathcal{M}_3$ because it has zero model error and the lowest training error.
- ☐ $\mathcal{M}_3$ because one should relax bias until the model error is exactly zero.
- ☐ $\mathcal{M}_3$ because it gives the smallest sum of estimation error and model error among the three.

31. What does "linear modeling" mean in the feature-map framework? 1 Point

- ☐ Predictors are linear combinations of chosen basis functions: $f_w(x) = w^T \phi(x)$.
- ☐ Predictors must be linear functions of the raw input coordinates.
- ☐ The output labels must be linear functions of the sample index.
- ☐ The feature map must be the identity map.

32. For two-dimensional input $x = (x_1, x_2)$, which feature map represents all polynomial functions of total degree at most 2? 1 Point

- ☐ $\phi(x) = (x_1, x_2)$.
- ☐ $\phi(x) = (1, x_1, x_2, x_1^2, x_1 x_2, x_2^2)$.
- ☐ $\phi(x) = (1, x_1^2, x_2^2)$ only.
- ☐ $\phi(x) = (x_1, x_2)$ only.

25. As model complexity increases in an underparameterized regime, which qualitative behavior best captures the estimation/model-error tradeoff?

- ☐ Both estimation error and model error must decrease.
- ☐ Both estimation error and model error must increase.
- ☐ Model error often decreases because the class is richer, while estimation error often increases because more parameters must be learned from finite data.
- ☐ Training error and true risk must have identical curves as functions of model complexity.

26. Using

$$R(w^*) - R(f^*) = (\mathbb{E}[N])^2 + \text{Var}(\mathbb{E}[N \mid \mathcal{X}]),$$

which condition is necessary and sufficient for zero model error?

- ☐ $\mathbb{E}[N] = 0$ and $\mathbb{E}[N \mid \mathcal{X}]$ is almost surely constant, equivalently $\mathbb{E}[N \mid \mathcal{X}] = 0$ almost surely.
- ☐ $\mathbb{E}[\mathcal{X}N] = 0$.
- ☐ N is Gaussian.
- ☐ $\mathbb{E}[N^2] = 0$.

27. Why does the average loss on an independent test set estimate the true risk of a learned predictor?

- ☐ Because the test loss is always smaller than the training loss.
- ☐ Because the predictor has zero empirical risk on the training set.
- ☐ Because the test set contains the same examples as the training set.
- ☐ Because, conditional on the trained predictor, independent test losses are iid samples from the loss distribution, so their average concentrates around its expectation.

28. A practitioner splits data into training and test sets. They train polynomial models of degrees 1 through 20, choose the degree with the lowest test error, and then report that same test error as the final generalization estimate. What is the main problem?

- ☐ The test set is invalid because polynomial models cannot be evaluated on it.
- ☐ The test set has been used for model selection, so the reported estimate is optimistically biased.
- ☐ The training set should have been used only after the test set.
- ☐ The procedure is valid because all models were trained only on the training set.

21. Why is supervised learning not merely denoising in the regression setting

$$Y = w^*{}^\top X + N, \quad \mathbb{E}[XN] = 0?$$

- ☐ Denoising focuses on recovering w^* , while supervised prediction targets $\mathbb{E}[Y \mid X = x] = w^*{}^\top x + \mathbb{E}[N \mid X = x]$.
- ☐ Supervised learning ignores the conditional distribution of Y .
- ☐ Denoising is possible only for classification problems.
- ☐ Supervised learning always assumes N is independent Gaussian noise.

22. Suppose

$$Y = 2X + N, \quad \mathbb{E}[XN] = 0, \quad \mathbb{E}[N \mid X = x] = x^2.$$

What is the square-loss Bayes predictor?

- ☐ $2x$.
- ☐ x^2 .
- ☐ $2x + x^2$.
- ☐ $2x + \mathbb{E}[N]$.

23. Let f^* be the unrestricted square-loss Bayes predictor, w^* the best predictor in a restricted linear class, and $\hat{w}_m$ the ERM solution in that class. Which decomposition is correct?

- ☐ $R(\hat{w}_m) - R(f^*) = (R(\hat{w}_m) - R(w^*)) + (R(w^*) - R(f^*))$.
- ☐ $R(\hat{w}_m) - R(f^*) = (R(w^*) - R(\hat{w}_m)) + (R(f^*) - R(w^*))$.
- ☐ $R(\hat{w}_m) - R(f^*) = R(\hat{w}_m) - R(w^*)$ always.
- ☐ $R(\hat{w}_m) - R(f^*) = R(w^*) - R(f^*)$ always.

24. For the decomposition $Y = w^*{}^\top X + N$ with $\mathbb{E}[XN] = 0$, the model error of the best linear predictor relative to the unrestricted square-loss Bayes predictor can be written as $R(w^*) - R(f^*)$.

Which expression is correct?

- ☐ $\mathbb{E}[\text{Var}(N \mid X)]$.
- ☐ $(\mathbb{E}[N])^2 + \text{Var}(\mathbb{E}[N \mid X])$.
- ☐ $\mathbb{E}[N^2] + \mathbb{E}[\text{Var}(N \mid X)]$.
- ☐ $\mathbb{E}[N^2]$ in all cases.

17. In underparameterized linear-regression estimation, a typical expected excess-risk behavior is

$$\mathbb{E}[R(\hat{w}_m) - R(w^*)] \approx \mathbb{E}[N^2] \frac{1}{m-n}, \quad m > n,$$

where n is the number of parameters and m is the number of samples. Which qualitative conclusion is best supported by this expression?

- ☐ Increasing n always decreases estimation error because the model is more expressive.
- ☐ More noise and more parameters make recovery harder, while increasing m helps; the expression also warns of instability when m is close to n .
- ☐ Once $m > n$, estimation error is independent of sample size.
- ☐ Estimation error is controlled only by the Bayes optimal function $\mathbb{E}[Y \mid X]$.

18. Let $u \in \mathbb{R}^d$ be a zero-mean random vector with covariance $C = \mathbb{E}[uu^\top]$, and let M be a deterministic $d \times d$ matrix. Which identity is the trace trick used in linear-regression estimation-error calculations?

- ☐ $\mathbb{E}[u^\top M u] = \text{tr}(M) \text{tr}(C)$.
- ☐ $\mathbb{E}[u^\top M u] = \text{tr}(MC)$.
- ☐ $\mathbb{E}[u^\top M u] = u^\top \mathbb{E}[M] u$.
- ☐ $\mathbb{E}[u^\top M u] = \text{tr}(M^{-1}C)$.

19. Let $\hat{\Sigma} = \frac{1}{n} \sum_i x_i x_i^\top$ and $\Sigma = \mathbb{E}[XX^\top] \succ 0$. Suppose $\hat{\Sigma}$ is positive definite. The estimation-error analysis contains the term $\mathbb{E}[\text{tr}(\hat{\Sigma}^{-1}\Sigma)]$.

Which Jensen argument gives the standard lower bound?

- ☐ Since inversion is linear, $\mathbb{E}[\hat{\Sigma}^{-1}] = (\mathbb{E}[\hat{\Sigma}])^{-1}$, so the term equals d exactly.
- ☐ Since $A \mapsto \text{tr}(A^{-1}\Sigma)$ is concave, Jensen gives an upper bound d .
- ☐ Jensen cannot be applied to matrices, so no lower bound is possible.
- ☐ Since $A \mapsto \text{tr}(A^{-1}\Sigma)$ is convex on positive definite matrices, $\mathbb{E}[\text{tr}(\hat{\Sigma}^{-1}\Sigma)] \geq \text{tr}((\mathbb{E}\hat{\Sigma})^{-1}\Sigma) = d$.

20. In one estimation-error analysis for empirical linear regression, the expected excess risk contains the term

$$\frac{\mathbb{E}[N^2]}{m} \mathbb{E}\left[\text{tr}\left(\left(\frac{1}{m} \sum_{i=1}^m x_i x_i^\top\right)^{-1} \mathbb{E}[XX^\top]\right)\right].$$

Jensen's inequality gives a lower bound n for the trace term. For Gaussian X , the corresponding expression behaves like

$$\frac{nm}{m-n-1},$$

and in a high-dimensional asymptotic picture it behaves like

$$\frac{nm}{m-n}.$$

Which interpretation is most consistent with these statements?

- ☐ The trace term is exactly n for every finite sample, so the inverse empirical correlation matrix causes no additional difficulty.
- ☐ Jensen's inequality shows that the expected estimation error is at most $\mathbb{E}[N^2] n/m$.
- ☐ The Gaussian and asymptotic expressions show that estimation error vanishes as m approaches n from above.
- ☐ The inverse empirical correlation matrix can be unstable at finite sample size; the estimation error is at least of order $\mathbb{E}[N^2] n/m$ and can behave like $\mathbb{E}[N^2] n/(m-n)$, becoming large near $m \approx n$.

13. Which equation is obtained by setting the gradient of

$$\frac{1}{m} \sum_{i=1}^m (w^\top x_i - y_i)^2$$

to zero?

- ☐ $\left(\frac{1}{m} \sum_{i=1}^m x_i x_i^\top\right) \hat{w}_m = \frac{1}{m} \sum_{i=1}^m x_i y_i.$
- ☐ $\left(\frac{1}{m} \sum_{i=1}^m x_i^\top x_i\right) \hat{w}_m = \frac{1}{m} \sum_{i=1}^m y_i.$
- ☐ $\hat{w}_m = \frac{1}{m} \sum_{i=1}^m x_i y_i$ without any matrix inverse or linear system.
- ☐ $\sum_{i=1}^m (w^\top x_i - y_i)^2 = 0.$

14. Let

$$A = \mathbb{E}[XX^\top], \quad b = \mathbb{E}[XY],$$

so that the population linear-regression parameter satisfies

$$Aw^* = b.$$

The empirical solution satisfies

$$\hat{A}_m \hat{w}_m = \hat{b}_m, \quad \hat{A}_m = \frac{1}{m} \sum_{i=1}^m x_i x_i^\top, \quad \hat{b}_m = \frac{1}{m} \sum_{i=1}^m x_i y_i.$$

Which statement best captures the law-of-large-numbers intuition for why $\hat{w}_m$ should approach w^* when there are enough samples and the correlation matrices are not singular?

- ☐ Convergence of $\hat{b}_m$ alone is enough; the empirical correlation matrix is irrelevant.
- ☐ Singular empirical correlation matrices always become invertible just by taking a formal limit.
- ☐ The empirical quantities $\hat{A}_m$ and $\hat{b}_m$ are sample averages of the corresponding population quantities, so they approach A and b ; solving a nearby well-behaved linear system should give a parameter close to w^* .
- ☐ $\hat{A}_m = A$ and $\hat{b}_m = b$ for every finite training set.

15. Which comparison between the population and empirical linear-regression solutions is most accurate?

- ☐ w^* and $\hat{w}_m$ are both deterministic once the model class is chosen.
- ☐ $\hat{w}_m$ is the Bayes optimal predictor, while w^* is the empirical minimizer.
- ☐ w^* uses the training-set distribution, while $\hat{w}_m$ uses the unknown population distribution.
- ☐ w^* depends on population moments such as $\mathbb{E}[XX^\top]$, whereas $\hat{w}_m$ depends on random sample moments and is therefore itself random.

16. Assume the data satisfy

$$y_i = w^{*\top} x_i + n_i,$$

and define $\hat{\Sigma} = \frac{1}{m} \sum_i x_i x_i^\top$. If $\hat{\Sigma}$ is invertible, which expression correctly gives the parameter error of ERM?

- ☐ $\hat{w}_m - w^* = \hat{\Sigma}^{-1} \left(\frac{1}{m} \sum_{i=1}^m x_i n_i \right)$
- ☐ $\hat{w}_m - w^* = \Sigma^{-1} \left(\frac{1}{m} \sum_{i=1}^m x_i n_i \right)$, where $\Sigma = \mathbb{E}[XX^\top]$, exactly for every sample.
- ☐ $\hat{w}_m - w^* = \frac{1}{m} \sum_{i=1}^m n_i.$
- ☐ $\hat{w}_m - w^* = 0$ whenever $\mathbb{E}[XN] = 0.$

9. Consider the Hilbert space of square-integrable random variables with inner product $\langle A, B \rangle = \mathbb{E}[AB]$. Linear regression projects Y onto $\text{span}(X_1, \dots, X_d)$. If $Y = w^*{}^\top X + N$, what does the projection theorem imply?

- ☐ N is independent of X .
- ☐ $N = 0$ almost surely.
- ☐ $\mathbb{E}[X_i N] = 0$ for every i , equivalently $\mathbb{E}[XN] = 0$.
- ☐ $\mathbb{E}[N | X] = 0$ always.

10. How are the gradient-based normal equations and the projection-theorem derivation of linear regression related?

- ☐ Both impose that the residual is orthogonal to every feature direction; one expresses this as zero gradient, the other as projection geometry.
- ☐ The gradient derivation applies only to empirical risk, whereas the projection derivation applies only to population risk.
- ☐ The projection derivation gives a different solution unless X and N are independent.
- ☐ The gradient derivation minimizes model error, whereas the projection derivation minimizes estimation error.

11. For linear regression, the empirical risk objective is

$$\hat{R}_m(w) = \frac{1}{m} \sum_{i=1}^m (w^\top x_i - y_i)^2.$$

Which interpretation is most accurate?

- ☐ ERM is unrelated to risk minimization because it uses finite sums.
- ☐ ERM assumes the true distribution is Gaussian.
- ☐ ERM minimizes the population risk but only after replacing Y by its conditional expectation.
- ☐ ERM is risk minimization under the empirical distribution that assigns mass $1/m$ to each observed sample.

12. In empirical-risk minimization for linear regression, the empirical correlation matrix is

$$\hat{\Sigma}_m = \frac{1}{m} \sum_{i=1}^m x_i x_i^\top,$$

where each $x_i \in \mathbb{R}^n$. Why is the condition $m > n$ natural in an underparameterized derivation that uses $\hat{\Sigma}_m^{-1}$?

- ☐ If $m < n$, then $\hat{\Sigma}_m$ is automatically positive definite because there are fewer samples.
- ☐ Each matrix $x_i x_i^\top$ has rank at most 1, so $\hat{\Sigma}_m$ has rank at most m ; hence $m < n$ makes $\hat{\Sigma}_m$ singular, and even when $m \geq n$, the samples must span all feature directions.
- ☐ Invertibility of $\hat{\Sigma}_m$ depends only on the noise energy $\mathbb{E}[N^2]$, not on the input samples.
- ☐ Jensen's inequality implies $\hat{\Sigma}_m$ is invertible for every finite sample.

5. Suppose X is a real-valued random variable with $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] > 0$, and $\mathbb{E}[X^3] = 0$. Let $Y = X^2$. Consider linear predictors of the form $f_w(X) = wX$, with no intercept. Which statement is correct?

- ☐ The full Bayes predictor and the restricted linear Bayes predictor are both X^2 .
- ☐ The best linear predictor is 0, while the full Bayes predictor is X^2 .
- ☐ The best linear predictor is X^2 because Y is deterministic given X .
- ☐ The best linear predictor does not exist because Y is nonlinear in X .

6. Let

$$R(w) = \mathbb{E}[(w^\top X - Y)^2], \quad \Sigma = \mathbb{E}[XX^\top], \quad b = \mathbb{E}[XY],$$

where $\Sigma \succ 0$. Which derivation correctly gives the population linear-regression solution?

- ☐ Since $R(w)$ is a square, the minimizer must satisfy $w^\top X = Y$ almost surely.
- ☐ Since Σ is positive definite, the minimizer is $w^* = \Sigma b$.
- ☐ Expanding gives $R(w) = w^\top \Sigma w + 2w^\top b + \mathbb{E}[Y^2]$, so $w^* = -\Sigma^{-1}b$.
- ☐ Expanding gives $R(w) = w^\top \Sigma w - 2w^\top b + \mathbb{E}[Y^2]$, so $\nabla R(w) = 2\Sigma w - 2b$ and $w^* = \Sigma^{-1}b$.

7. In the linear model class $f_w(x) = w^\top x$, what is the role of w ?

- ☐ w is an additional random input drawn from the data distribution.
- ☐ w is the observed label vector.
- ☐ w is a parameter vector indexing a member of the restricted function class.
- ☐ w is the residual noise after projection.

8. Suppose features have been whitened so that $\mathbb{E}[\tilde{X}\tilde{X}^\top] = I$. For the linear predictor $w^\top \tilde{X}$, what is the population optimal coefficient vector under square loss?

- ☐ $w^* = \mathbb{E}[\tilde{X}\tilde{X}^\top Y]$.
- ☐ $w^* = \mathbb{E}[\tilde{X}Y]$.
- ☐ $w^* = \mathbb{E}[Y]\mathbb{E}[\tilde{X}]$.
- ☐ $w^* = \mathbb{E}[\tilde{X}]^{-1}\mathbb{E}[Y]$.

1. A learning algorithm minimizes empirical risk over all measurable functions $f: \mathcal{X} \rightarrow \mathcal{Y}$ on a finite training set. Which statement best explains why an inductive bias or restricted model class is essential?

1 Point

- ☐ With enough training data, unrestricted ERM always converges uniformly over all measurable functions.
- ☐ A finite sample constrains behavior only on observed points; without restrictions, a function can interpolate the sample while behaving arbitrarily elsewhere.
- ☐ Inductive bias is needed only because stochastic gradient descent requires differentiability.
- ☐ If the empirical risk is zero, the expected risk is necessarily zero by the law of large numbers.

2. For a regression problem with square loss, suppose that conditional on $X = x$, the response is

1 Point

$$Y = \begin{cases} x + 1, & \text{with probability } 0.9, \\ x - 9, & \text{with probability } 0.1. \end{cases}$$

What is the Bayes optimal prediction at input x ?

- ☐ $x + 1$, because it is the most likely value of Y .
- ☐ $x - 9$, because the loss can be large when the rare event occurs.
- ☐ x , because the square-loss Bayes predictor is $\mathbb{E}[Y|X = x]$.
- ☐ Any value between $x - 9$ and $x + 1$, because both responses are possible.

3. Let $\mu(x) = \mathbb{E}[Y|X = x]$. Which identity gives the essential derivation of the square-loss Bayes predictor?

1 Point

- ☐ $\mathbb{E}[(e - Y)^2|X = x] = \mathbb{E}[Y^2|X = x] - e^2$
- ☐ $\mathbb{E}[(e - Y)^2|X = x] = \text{Var}(Y|X = x) - 2e\mu(x)$.
- ☐ $\mathbb{E}[(e - Y)^2|X = x] = (e - \mu(x))^2 + \text{Var}(Y|X = x)$.
- ☐ $\mathbb{E}[(e - Y)^2|X = x] = (e - \mu(x))^2 + \text{Var}(Y|X = x)$.

4. Let $X \in \mathbb{R}^d$ be a random vector with finite second moment and define $\mathcal{D} = \mathbb{E}[XX^T]$. Which argument correctly proves that $\mathcal{D}$ is positive semidefinite?

1 Point

- ☐ For every $v \in \mathbb{R}^d$, $v^T \mathcal{D} v = \mathbb{E}[(v^T X)^2] \geq 0$.
- ☐ Since every entry of XX^T is nonnegative, every entry of $\mathcal{D}$ is nonnegative.
- ☐ Since $\mathcal{D}$ is an expectation, it must be invertible.
- ☐ Since X is random, $\mathcal{D}$ has strictly positive eigenvalues.